

Detecting Cloned Voices on a Phone Call

1.23% on their own benchmark. 37% on someone else's.

We are building one that closes that gap.

Ninety seconds of audio is enough.

- A voice from a podcast or a voicemail.
- **Cloned in minutes.** No GPU farm, no expertise.
- A call to a parent, an employee, or a bank.

THE QUESTION

Can a machine tell that it is synthetic?

Excellent numbers, on **one dataset**.

BEST REPORTED

1.23%

RawGAT-ST on ASVspoof 2019 LA

CHALLENGES RUN

4

ASVspoof, biennial since 2015

ARCHITECTURES COMPARED

12

One protocol, raw waveform wins

THE CATCH

Every number here is measured on the corpus the model trained on.

Move it off its own dataset and it **stops working**.

TRAINED ON ASVSP00F 2019, TESTED ON:

its own test set

1.23%

another corpus

In-the-Wild

37.15%

2026 cloned voices

over a real VoIP call

50.28%

COIN FLIP

same test, retrained

on modern voices

5.81% so it can be fixed

equal error rate, lower is better

0% 15% 30% 45% 60%

A 36-point gap. Closing it is the job.

It learns the **dataset**, not the voice.

SILENCE ALONE

85%

accuracy from the length of the opening
silence

ATTACK ON SILENCE ONLY

82.2%

success without ever touching the voice

WHERE IT LOOKS

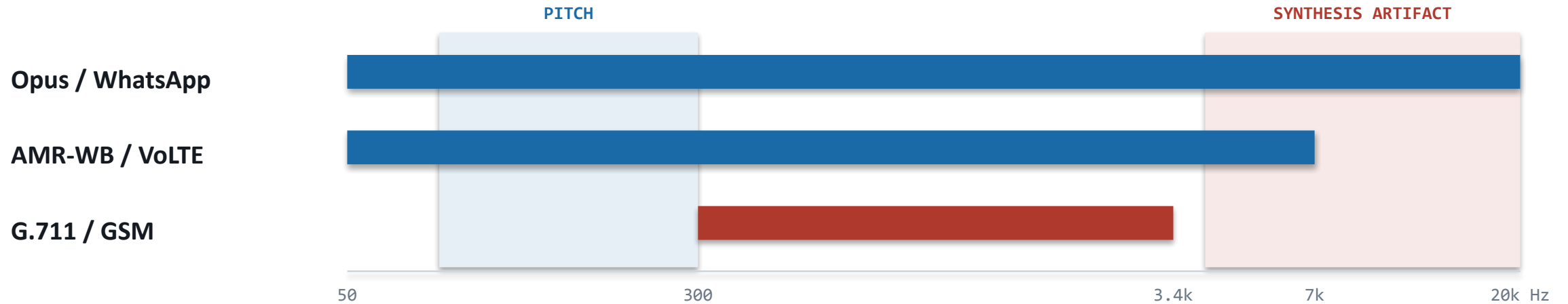
0.5-0.6 kHz

the gaps between words, one narrow
band

WHICH TELLS US WHAT TO FIX

It wins on what belongs to the corpus, not the speech. **All of it can be taken away in training.**

What survives a phone line?



GOOD NEWS

The phone keeps the pitch band. **The pitch band alone reaches 1.15% EER.**

How you build a detector that travels.

DESIGN DECISION	EVIDENCE
Codec and channel augmentation	every top-5 system in 2021 used it
Raw waveform or CQT, never mel	mel is 37% worse
Silence masking, VAD-trimmed training	removes the cue worth 85%
Low-pass to the telephone band	25% better under codec shift
SSL front-end (wav2vec2)	1.89% to 11.6% under attack, but heavy
One-class objective	learns real speech, not known attacks

None of these is new on its own. The work is making them hold against both shifts at once.

The only number that matters is the **gap**.

TRAIN

ASVspoof 2019 LA, clean.

One attack family held out.

TEST

ASVspoof 2021 LA for the channel

In-the-Wild for the corpus

OUR HEADLINE METRIC

gap = out-of-domain EER - in-domain EER

Reported per codec and per corpus. Never averaged together.

An app on the callee's phone.

WHAT THE SETTING GIVES US

Seconds, not milliseconds.

False alarms cost more than misses.

WHAT WE MUST NOT HIDE

Android will not give us the call audio.

So we capture through the room: **4.7% to 18.2%**

SO WE TRAIN AGAINST IT TOO

Re-capture is a third shift, and it is the one a real deployment forces on us.

One window is not a decision.

THE TRAP

5% error per window, 2-second windows, and a 60-second call false-alarms **79% of the time.**

HOW WE DECIDE

Add the evidence up

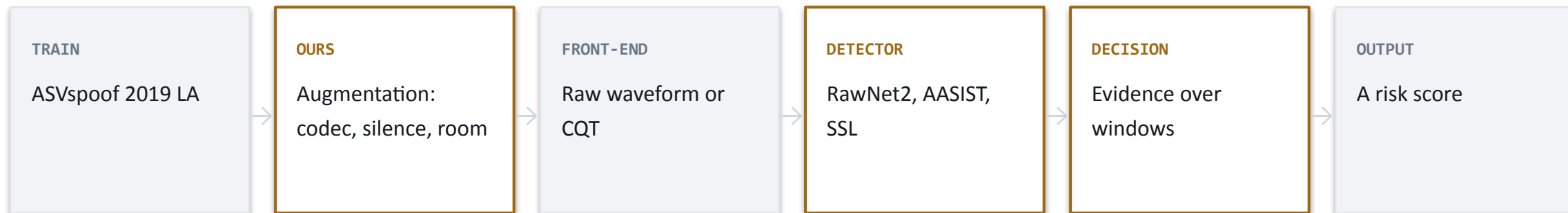
Alarm when the total crosses a boundary, not when one window does.

WHAT WE REPORT

Per call-minute

False alarms, detection rate, time to detection.

What we will build.



WHAT WE SWEEP

Seven codecs plus **AMR**, which no ASVspoof edition covers.

Window length: full, 4 s, 2 s, 1 s.

NEVER IN TRAINING

In-the-Wild, the held-out attacks, and a small probe of modern cloned voices.

Twelve weeks.



What success looks like.

THE TARGET

Out-of-domain EER **well below 34%.**

A gap we **reduced, and can explain.**

A decision **in seconds, on the phone.**

WHAT WE WILL NOT DO

Quote one in-domain number.

Average across corpora or codecs.

Say the problem is solved.

THE POSITION WE WILL DEFEND

A detector that travels is worth more than a detector that scores well.